



COMPUTER NETWORKS
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Project Report:

Smart Network Health Analyzer



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1. Introduction

The **Smart Network Health Analyzer** is a Python-based desktop application used for analyzing the quality and performance of a computer network for a given website or domain name, for example, google.com. The project helps an end-user understand how healthy their internet connection is by measuring important network parameters: latency, jitter, DNS resolution, and an overall network health score.

“This project is very helpful for beginners, as it depicts several networking concepts in the real world through a very friendly and simple graphical interface.”

2. Project Objectives

The main goals of this project are:

- ⇒ To study network performance for a given domain name
- ⇒ In order to measure the TCP latency and jitter.
- ⇒ To check DNS resolution performance
- ⇒ To compute a network health score
- ⇒ To present findings in a graphical and easy-to-comprehend format
- ⇒ To provide a beginner-friendly tool for understanding networking concepts.

3. Scope of the Project

The project deals with network performance analysis at the level of the user. This task does not require special hardware or administration privileges.

Scope:

- ⇒ Domain name analysis: e.g. google.com
- ⇒ TCP-based latency measurement
- ⇒ Basic network quality evaluation
- ⇒ Visualization of results by graphical means

4. Tools and Technologies Used

4.1 Programming Language

- ⇒ Python 3-used for the implementation of network logic and GUI

4.2 Libraries and Modules

- ⇒ **socket** - for DNS resolution and TCP connection
- ⇒ **time** – to measure delays
- ⇒ **statistics** - to calculate average and jitter
- ⇒ **tkinter** - for developing the GUI
- ⇒ **matplotlib** – for plotting latency graphs

4.3 Platform

- ⇒ Operating System of Windows

5. System Architecture

The system follows a simple architecture:

- ⇒ User inputs a domain name.
- ⇒ GUI sends input to backend logic.
- ⇒ DNS resolution is carried out.
- ⇒ TCP latency tests are performed.
- ⇒ Jitter and averages are calculated
- ⇒ Network health score is generated.
- ⇒ Results are displayed on GUI and graph

6. Explanation of the Working Project (Backend)

6.1 User Input

The user simply provides a domain name, for example, google.com, in the GUI and then clicks on the 'Analyze Network' button.

6.2 DNS Resolution

DNS converts the domain name into an IP address. This is how the system will communicate with the target server.

6.3 TCP Latency Measurement

The script establishes multiple TCP connections to the targeted IP, and the round-trip time for every connection is measured in milliseconds.

6.4 Calculation of Average Latency

This will then be averaged to compute the TCP average latency.

6.5 Jitter Calculation

Jitter is determined by the amount of change between successive latency measurements.

6.6 Network Health Score

Calculate the health score based on latency, jitter, and DNS success from 0 to 100.

6.7 Visualizing Results

- ⇒ Numerical results are shown on the GUI.
- ⇒ Latency variations of the A Bar graph.
- ⇒ Network quality is categorized as Good, Average, or Poor.

7. Output Description

- ⇒ The **output** of the project includes:
- ⇒ TCP Average Latency (ms)
- ⇒ DNS Average Latency (ms)
- ⇒ Jitter (ms)
- ⇒ Network Health Score
- ⇒ Network Quality Message
- ⇒ Latency Graph

8. Applications of the Project

- ⇒ Testing the quality of the internet

- ⇒ Educational tool for networking students
- ⇒ Basic troubleshooting of network problems
- ⇒ Understand real-world TCP and DNS behavior

9. Benefits

- ⇒ Beginner-friendly
- ⇒ Real-time network analysis
- ⇒ Graphical visualization
- ⇒ Single-file implementation in pure Python
- ⇒ No external hardware required

10. Limitations

- ⇒ Requires an internet connection
- ⇒ Limited by basic metrics only
- ⇒ Results would depend upon network conditions.

11. Future Improvements

- ⇒ Add packet loss detection
- ⇒ Add bandwidth speed testing
- ⇒ Export report to PDF
- ⇒ Improve scoring algorithm
- ⇒ Add multi-server comparison

12. Conclusion

The Smart Network Health Analyzer is quite able to demonstrate the use of basic concepts in networking, such as DNS resolution, TCP latency, and jitter in a pretty workable application. The project is at a beginner level, thus forming a very good foundation for learning network performance analysis.

